

# Alejandro Gomez De Mendieta

[contact@agomezdemen.com](mailto:contact@agomezdemen.com) | [agomezdemen.com](https://agomezdemen.com) | [LinkedIn](#) | [GitHub](#)

## EDUCATION

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### The University of Texas at Dallas

*Bachelor of Science in Computer Science, Minor in Statistics*

Aug. 2025 – Dec. 2027

*Richardson, TX*

### Austin Community College

*Associate of Science in Computer Science; Associate of Science in Mathematics*

Aug. 2023 – Aug. 2025

*Austin, TX*

## EXPERIENCE

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### Software Engineer (Part-time)

*Ohel Technologies*

Jan. 2022 – Jul. 2025

*Austin, TX*

- Automated internal data and IT workflows for a five-person team and documented them in five SOPs, reducing recurring manual work by approximately five hours per week.
- Engineered biweekly ingestion pipelines integrating five external APIs with Elasticsearch, loading 100,000+ records in three months to support an AI application.
- Designed quota-aware ingestion logic around third-party rate limits and monthly usage caps to maintain reliable data collection.

### Computer Science II Course Grader

*University of Texas at Dallas*

Jan. 2026 – May 2026

*Richardson, TX*

- Graded approximately 30 Java submissions per week for 71 students across two sections, maintaining a typical three-day turnaround against a seven-day policy.
- Provided rubric-aligned feedback on correctness, edge cases, code quality, and specification compliance to help students diagnose logic errors and improve implementations.
- Maintained shared notes on recurring mistakes and ambiguous edge cases to improve grading consistency across two graders.

## PROJECTS

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### Modern C++ HTTP Server | C++23, Linux, POSIX sockets, CMake

May 2026 – Present

- Built a from-scratch HTTP/1.1 server in C++23 using Linux/POSIX sockets, move-only RAII file descriptor ownership, layered TCP/HTTP abstractions, method/path-based routing, and incremental request parsing.
- Implemented non-blocking event-driven I/O with Linux epoll, managing per-connection state across fragmented reads and writes and scaling a single-threaded server to thousands of concurrent connections.
- Built a performance-testing framework using custom nanosecond-resolution microbenchmarks, wrk load testing, and Linux perf, measuring throughput and mean/p95/p99 latency across workloads up to 100,000 iterations and increasing connection counts.
- Profiled parser workloads with Linux perf, traced ~85% of sampled self CPU to repeated buffer movement in `memmove`, and redesigned parsing around a cursor-based buffer, reducing latency by ~10–17% across affected workloads; validated correctness with 121 Catch2 unit and process-level TCP integration tests.

### Homefull, HackUTD | Python, NVIDIA Nemotron, DigitalOcean

Nov. 8–9, 2025

- Deployed NVIDIA Nemotron on a DigitalOcean GPU droplet and built a Python backend that transformed crawled webpage content into structured JSON for local resource and event discovery.
- Implemented URL fetching, HTML parsing, event extraction, geocoding, and deduplication across the top five search results and pages crawled within each domain.
- Integrated an end-to-end discovery pipeline that returned event time and location information to the mobile application.

## TECHNICAL SKILLS

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**Languages:** C++, C, Python, Java, R

**Systems and Tools:** Linux/UNIX, POSIX sockets, TCP/IP, CMake, Git, CTest, GitHub Actions, Linux perf, Docker, Redis, Elasticsearch

**Libraries:** Catch2, nlohmann/json